VB_R_assignment

2026-03-15

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Part I. Data Inspection and Processing

Input and output paths

```r
dir.create("02_output_files", showWarnings = FALSE)

geno_file <- "00_data_files/fang_et_al_genotypes.txt"
pos_file  <- "00_data_files/snp_position.txt"
out_dir   <- "02_output_files"

dir.create(out_dir, showWarnings = FALSE, recursive = TRUE)
dir.create(file.path(out_dir, "maize"), showWarnings = FALSE, recursive = TRUE)
dir.create(file.path(out_dir, "teosinte"), showWarnings = FALSE, recursive = TRUE)
```

Read input files

```r
geno <- readr::read_tsv(
  geno_file,
  col_types = cols(.default = col_character()),
  progress = FALSE
)

pos_raw <- readr::read_tsv(
  pos_file,
  col_types = cols(.default = col_character()),
  progress = FALSE
)
```

Basic inspection

```r
cat("Genotype file:\n")
```

```
## Genotype file:
```

```r
cat("  Rows:", nrow(geno), "\n")
```

```
##   Rows: 2782
```

```r
cat("  Columns:", ncol(geno), "\n")
```

```
##   Columns: 986
```

```r
cat("  Size (bytes):", file.info(geno_file)$size, "\n\n")
```

```
##   Size (bytes): 11051939
```

```r
cat("Position file:\n")
```

```
## Position file:
```

```r
cat("  Rows:", nrow(pos_raw), "\n")
```

```
##   Rows: 983
```

```r
cat("  Columns:", ncol(pos_raw), "\n")
```

```
##   Columns: 15
```

```r
cat("  Size (bytes):", file.info(pos_file)$size, "\n\n")
```

```
##   Size (bytes): 82763
```

Preview files geno = genotype file, pos_raw = snp_position file

```r
head(geno)
```

```
## # A tibble: 6 x 986
##   Sample_ID JG_OTU   Group abph1.20 abph1.22 ae1.3 ae1.4 ae1.5 an1.4 ba1.6 ba1.9
##   <chr>     <chr>    <chr> <chr>    <chr>    <chr> <chr> <chr> <chr> <chr> <chr>
## 1 SL-15     T-aust-1 TRIPS ?/?      ?/?      T/T   G/G   T/T   C/C   ?/?   G/G
## 2 SL-16     T-aust-2 TRIPS ?/?      ?/?      T/T   ?/?   T/T   C/C   A/G   G/G
## 3 SL-11     T-brav-1 TRIPS ?/?      ?/?      T/T   G/G   T/T   ?/?   G/G   G/G
## 4 SL-12     T-brav-2 TRIPS ?/?      ?/?      T/T   G/G   T/T   C/C   G/G   G/G
## 5 SL-18     T-cund   TRIPS ?/?      ?/?      T/T   G/G   T/T   C/C   ?/?   G/G
## 6 SL-2      T-dact-1 TRIPS ?/?      ?/?      T/T   G/G   T/T   C/C   A/G   G/G
## # i 975 more variables: bt2.5 <chr>, bt2.7 <chr>, bt2.8 <chr>, Fea2.1 <chr>,
## #   Fea2.5 <chr>, id1.3 <chr>, lg2.11 <chr>, lg2.2 <chr>, pbf1.1 <chr>,
## #   pbf1.2 <chr>, pbf1.3 <chr>, pbf1.5 <chr>, pbf1.6 <chr>, pbf1.7 <chr>,
## #   pbf1.8 <chr>, PZA00003.11 <chr>, PZA00004.2 <chr>, PZA00005.8 <chr>,
## #   PZA00005.9 <chr>, PZA00006.13 <chr>, PZA00006.14 <chr>, PZA00008.1 <chr>,
## #   PZA00010.5 <chr>, PZA00013.10 <chr>, PZA00013.11 <chr>, PZA00013.9 <chr>,
## #   PZA00015.4 <chr>, PZA00017.1 <chr>, PZA00018.5 <chr>, ...
```

```r
tail(geno)
```

```
## # A tibble: 6 x 986
##   Sample_ID JG_OTU   Group abph1.20 abph1.22 ae1.3 ae1.4 ae1.5 an1.4 ba1.6 ba1.9
##   <chr>     <chr>    <chr> <chr>    <chr>    <chr> <chr> <chr> <chr> <chr> <chr>
## 1 SYN262    Zmm-IL-~ ZMMIL C/C      A/A      T/T   G/G   C/C   C/C   G/G   G/G
## 2 S0398     Zmm-IL-~ ZMMIL G/G      A/A      T/T   G/G   C/C   C/C   G/G   G/G
## 3 S1636     Zmm-IL-~ ZMMIL G/G      A/A      T/T   G/G   C/C   C/C   G/G   G/G
## 4 CU0201    Zmm-IL-~ ZMMIL C/C      A/A      T/T   G/G   C/C   C/C   G/G   G/G
## 5 S0215     Zmm-IL-~ ZMMIL G/G      A/A      T/T   ?/?   C/C   C/C   G/G   G/G
## 6 CU0202    Zmm-IL-~ ZMMIL C/C      A/A      T/T   G/G   C/C   C/C   ?/?   G/G
## # i 975 more variables: bt2.5 <chr>, bt2.7 <chr>, bt2.8 <chr>, Fea2.1 <chr>,
## #   Fea2.5 <chr>, id1.3 <chr>, lg2.11 <chr>, lg2.2 <chr>, pbf1.1 <chr>,
## #   pbf1.2 <chr>, pbf1.3 <chr>, pbf1.5 <chr>, pbf1.6 <chr>, pbf1.7 <chr>,
## #   pbf1.8 <chr>, PZA00003.11 <chr>, PZA00004.2 <chr>, PZA00005.8 <chr>,
## #   PZA00005.9 <chr>, PZA00006.13 <chr>, PZA00006.14 <chr>, PZA00008.1 <chr>,
## #   PZA00010.5 <chr>, PZA00013.10 <chr>, PZA00013.11 <chr>, PZA00013.9 <chr>,
## #   PZA00015.4 <chr>, PZA00017.1 <chr>, PZA00018.5 <chr>, ...
```

```r
names(geno)
```

```
##   [1] "Sample_ID"      "JG_OTU"         "Group"          "abph1.20"
##   [5] "abph1.22"       "ae1.3"          "ae1.4"          "ae1.5"
##   [9] "an1.4"          "ba1.6"          "ba1.9"          "bt2.5"
```

```
##  [13] "bt2.7"          "bt2.8"          "Fea2.1"         "Fea2.5"
##  [17] "id1.3"          "lg2.11"         "lg2.2"          "pbf1.1"
##  [21] "pbf1.2"         "pbf1.3"         "pbf1.5"         "pbf1.6"
##  [25] "pbf1.7"         "pbf1.8"         "PZA00003.11"    "PZA00004.2"
##  [29] "PZA00005.8"     "PZA00005.9"     "PZA00006.13"    "PZA00006.14"
##  [33] "PZA00008.1"     "PZA00010.5"     "PZA00013.10"    "PZA00013.11"
##  [37] "PZA00013.9"     "PZA00015.4"     "PZA00017.1"     "PZA00018.5"
##  [41] "PZA00029.11"    "PZA00029.12"    "PZA00030.11"    "PZA00031.5"
##  [45] "PZA00041.3"     "PZA00042.2"     "PZA00042.5"     "PZA00043.7"
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##  [53] "PZA00051.2"     "PZA00058.5"     "PZA00058.6"     "PZA00060.2"
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##  [61] "PZA00078.2"     "PZA00079.1"     "PZA00081.17"    "PZA00084.2"
##  [65] "PZA00084.3"     "PZA00086.8"     "PZA00088.3"     "PZA00090.2"
##  [69] "PZA00092.1"     "PZA00092.5"     "PZA00093.2"     "PZA00096.26"
##  [73] "PZA00097.13"    "PZA00098.14"    "PZA00100.10"    "PZA00100.12"
##  [77] "PZA00100.14"    "PZA00100.9"     "PZA00103.20"    "PZA00106.9"
##  [81] "PZA00107.18"    "PZA00108.12"    "PZA00108.14"    "PZA00108.15"
##  [85] "PZA00109.3"     "PZA00109.5"     "PZA00111.2"     "PZA00111.4"
##  [89] "PZA00111.5"     "PZA00111.6"     "PZA00111.8"     "PZA00114.3"
##  [93] "PZA00116.2"     "PZA00119.4"     "PZA00120.4"     "PZA00123.1"
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## [433] "PZA00622.2"     "PZA00623.2"     "PZA00626.3"     "PZA00626.4"
## [437] "PZA00630.9"     "PZA00636.5"     "PZA00636.6"     "PZA00637.4"
## [441] "PZA00639.12"    "PZA00639.13"    "PZA00639.15"    "PZA00641.7"
```

```
## [445] "PZA00641.8"     "PZA00644.11"    "PZA00647.9"     "PZA00650.8"
## [449] "PZA00654.10"    "PZA00654.12"    "PZA00655.1"     "PZA00656.15"
## [453] "PZA00656.16"    "PZA00656.18"    "PZA00656.4"     "PZA00658.19"
## [457] "PZA00658.23"    "PZA00662.3"     "PZA00665.6"     "PZA00667.1"
## [461] "PZA00672.6"     "PZA00672.8"     "PZA00673.2"     "PZA00674.3"
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## [541] "PZA02853.10"    "PZA02853.7"     "PZA02856.1"     "PZA02862.3"
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## [557] "PZA02906.12"    "PZA02906.7"     "PZA02921.9"     "PZA02923.7"
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## [573] "PZA02949.26"    "PZA02952.10"    "PZA02954.2"     "PZA02955.3"
## [577] "PZA02958.17"    "PZA02959.7"     "PZA02961.1"     "PZA02962.13"
## [581] "PZA02963.5"     "PZA02966.11"    "PZA02968.4"     "PZA02969.11"
## [585] "PZA02970.9"     "PZA02972.1"     "PZA02982.5"     "PZA02982.6"
## [589] "PZA02983.38"    "PZA02984.7"     "PZA02988.2"     "PZA02993.5"
## [593] "PZA02997.16"    "PZA02997.19"    "PZA03001.15"    "PZA03001.18"
## [597] "PZA03001.9"     "PZA03009.5"     "PZA03009.6"     "PZA03009.7"
## [601] "PZA03009.8"     "PZA03011.6"     "PZA03012.10"    "PZA03013.7"
## [605] "PZA03013.8"     "PZA03014.10"    "PZA03014.21"    "PZA03014.24"
## [609] "PZA03017.10"    "PZA03017.11"    "PZA03024.16"    "PZA03024.18"
## [613] "PZA03024.7"     "PZA03028.5"     "PZA03032.16"    "PZA03034.1"
## [617] "PZA03035.5"     "PZA03037.8"     "PZA03037.9"     "PZA03041.8"
## [621] "PZA03042.1"     "PZA03042.5"     "PZA03046.2"     "PZA03046.3"
## [625] "PZA03047.12"    "PZA03047.20"    "PZA03047.22"    "PZA03048.16"
## [629] "PZA03048.17"    "PZA03049.23"    "PZA03051.1"     "PZA03051.3"
## [633] "PZA03052.15"    "PZA03054.3"     "PZA03054.5"     "PZA03058.17"
## [637] "PZA03062.15"    "PZA03062.7"     "PZA03063.17"    "PZA03063.18"
## [641] "PZA03064.6"     "PZA03067.17"    "PZA03067.20"    "PZA03068.11"
## [645] "PZA03068.13"    "PZA03069.6"     "PZA03073.23"    "PZA03073.24"
## [649] "PZA03074.24"    "PZA03078.33"    "PZA03081.1"     "PZA03081.10"
## [653] "PZA03081.11"    "PZA03081.13"    "PZA03081.6"     "PZA03083.7"
## [657] "PZA03089.12"    "PZA03090.31"    "PZA03092.7"     "PZA03094.18"
```

```
## [661] "PZA03094.6"     "PZA03095.1"     "PZA03095.2"     "PZA03095.3"
## [665] "PZA03097.4"     "PZA03097.7"     "PZA03097.9"     "PZA03102.10"
## [669] "PZA03102.2"     "PZA03102.9"     "PZA03137.1"     "PZA03172.2"
## [673] "PZA03223.3"     "PZA03258.2"     "PZA03283.2"     "PZA03284.3"
## [677] "PZA03290.1"     "PZA03290.2"     "PZA03295.4"     "PZA03296.6"
## [681] "PZA03296.7"     "PZA03298.1"     "PZA03298.2"     "PZA03301.2"
## [685] "PZA03301.4"     "PZA03302.1"     "PZA03305.6"     "PZA03305.7"
## [689] "PZA03311.2"     "PZA03311.3"     "PZA03311.4"     "PZA03311.5"
## [693] "PZA03312.1"     "PZA03312.2"     "PZA03316.2"     "PZA03317.1"
## [697] "PZA03319.3"     "PZA03319.4"     "PZA03320.3"     "PZA03320.4"
## [701] "PZA03328.5"     "PZA03329.1"     "PZA03329.2"     "PZA03333.3"
## [705] "PZA03335.2"     "PZA03335.3"     "PZA03337.1"     "PZA03338.5"
## [709] "PZA03340.2"     "PZA03342.2"     "PZA03344.4"     "PZA03344.5"
## [713] "PZA03344.6"     "PZA03345.1"     "PZA03345.2"     "PZA03345.4"
## [717] "PZA03347.1"     "PZA03348.1"     "PZA03349.1"     "PZA03349.9"
## [721] "PZA03767.1"     "PZA03767.4"     "PZA03767.5"     "PZA03773.2"
## [725] "PZA03773.3"     "PZA03774.1"     "PZA03774.10"    "PZA03774.2"
## [729] "PZA03774.4"     "PZA03774.5"     "PZA03774.6"     "PZA03774.8"
## [733] "PZA03774.9"     "PZA03775.1"     "PZA03775.11"    "PZA03775.2"
## [737] "PZA03775.3"     "PZA03775.4"     "PZA03775.6"     "PZA03775.7"
## [741] "PZA03775.8"     "PZA03775.9"     "PZA03781.1"     "PZA03781.2"
## [745] "PZA03781.3"     "PZA03781.4"     "PZA03781.5"     "PZA03781.6"
## [749] "PZA03781.7"     "PZA03781.8"     "PZA03782.1"     "PZA03782.3"
## [753] "PZA03786.1"     "PZA03786.2"     "PZA03789.1"     "PZA03789.2"
## [757] "PZA03789.4"     "PZB00011.4"     "PZB00011.5"     "PZB00041.2"
## [761] "PZB00041.4"     "PZB00049.2"     "PZB00049.4"     "PZB00049.7"
## [765] "PZB00055.1"     "PZB00060.4"     "PZB00062.6"     "PZB00062.7"
## [769] "PZB00062.8"     "PZB00067.2"     "PZB00067.3"     "PZB00067.4"
## [773] "PZB00067.5"     "PZB00078.1"     "PZB00081.2"     "PZB00081.4"
## [777] "PZB00081.5"     "PZB00081.7"     "PZB00092.1"     "PZB00092.4"
## [781] "PZB00093.3"     "PZB00093.4"     "PZB00093.6"     "PZB00096.2"
## [785] "PZB00096.3"     "PZB00136.3"     "PZB00140.1"     "PZB00145.2"
## [789] "PZB00149.2"     "PZB00149.4"     "PZB00153.1"     "PZB00153.2"
## [793] "PZB00153.3"     "PZB00153.5"     "PZB00160.1"     "PZB00160.2"
## [797] "PZB00160.4"     "PZB00165.2"     "PZB00165.6"     "PZB00169.4"
## [801] "PZB00169.6"     "PZB00175.1"     "PZB00175.2"     "PZB00175.3"
## [805] "PZB00175.4"     "PZB00175.5"     "PZB00180.1"     "PZB00180.2"
## [809] "PZB00183.3"     "PZB00188.6"     "PZB00207.3"     "PZB00221.3"
## [813] "PZB00221.8"     "PZB00229.3"     "PZB00232.1"     "PZB00232.2"
## [817] "PZB00232.4"     "PZB00232.5"     "PZB00379.3"     "PZB00379.4"
## [821] "PZB00379.5"     "PZB00393.7"     "PZB00409.3"     "PZB00416.2"
## [825] "PZB00416.5"     "PZB00454.2"     "PZB00454.3"     "PZB00454.4"
## [829] "PZB00454.5"     "PZB00498.2"     "PZB00498.4"     "PZB00598.1"
## [833] "PZB00598.2"     "PZB00603.3"     "PZB00603.4"     "PZB00603.5"
## [837] "PZB00607.2"     "PZB00761.1"     "PZB00761.2"     "PZB00849.2"
## [841] "PZB00849.3"     "PZB00849.4"     "PZB00859.1"     "PZB01109.2"
## [845] "PZB01109.3"     "PZB01110.1"     "PZB01110.2"     "PZB01110.3"
## [849] "PZB01111.6"     "PZB01111.7"     "PZB01111.8"     "PZB01112.3"
## [853] "PZB01112.4"     "PZB01112.5"     "PZB01112.6"     "PZB01113.4"
## [857] "PZB01114.1"     "PZB01114.3"     "PZB01115.1"     "PZB01115.5"
## [861] "PZB01115.6"     "PZB01116.2"     "PZB01221.1"     "PZB01222.1"
## [865] "PZB01222.3"     "PZB01223.3"     "PZB01223.4"     "PZB01223.7"
## [869] "PZB01225.1"     "PZB01225.2"     "PZB01225.4"     "PZB01228.1"
## [873] "PZB01228.3"     "PZB01228.4"     "PZB01233.2"     "PZB01233.3"
```

```
## [877] "PZB01238.5"     "PZB01238.6"     "PZB01427.1"     "PZB01427.3"
## [881] "PZB01463.2"     "PZB01463.3"     "PZB01463.4"     "PZD00003.1"
## [885] "PZD00003.3"     "PZD00007.1"     "PZD00008.3"     "PZD00011.1"
## [889] "PZD00011.3"     "PZD00011.4"     "PZD00012.1"     "PZD00012.2"
## [893] "PZD00012.3"     "PZD00012.4"     "PZD00012.5"     "PZD00013.3"
## [897] "PZD00013.4"     "PZD00014.3"     "PZD00017.1"     "PZD00019.1"
## [901] "PZD00020.2"     "PZD00020.3"     "PZD00020.4"     "PZD00020.6"
## [905] "PZD00021.2"     "PZD00021.4"     "PZD00021.5"     "PZD00022.1"
## [909] "PZD00022.3"     "PZD00022.4"     "PZD00024.2"     "PZD00025.1"
## [913] "PZD00025.2"     "PZD00030.1"     "PZD00030.4"     "PZD00030.5"
## [917] "PZD00030.6"     "PZD00034.3"     "PZD00043.1"     "PZD00043.2"
## [921] "PZD00043.3"     "PZD00043.4"     "PZD00044.2"     "PZD00044.3"
## [925] "PZD00044.4"     "PZD00045.1"     "PZD00045.2"     "PZD00045.3"
## [929] "PZD00045.4"     "PZD00049.3"     "PZD00049.4"     "PZD00049.5"
## [933] "PZD00051.1"     "PZD00052.3"     "PZD00052.4"     "PZD00062.2"
## [937] "PZD00066.1"     "PZD00067.1"     "PZD00067.2"     "PZD00067.3"
## [941] "PZD00068.1"     "PZD00069.2"     "PZD00069.3"     "PZD00069.4"
## [945] "PZD00069.5"     "PZD00073.1"     "PZD00073.2"     "PZD00073.6"
## [949] "PZD00074.1"     "PZD00075.1"     "PZD00075.2"     "PZD00076.1"
## [953] "PZD00076.2"     "PZD00076.4"     "PZD00077.10"    "PZD00077.5"
## [957] "PZD00077.7"     "PZD00077.8"     "PZD00078.2"     "Ra2_ORF.1"
## [961] "Ra2_ORF.2"      "Ra2_ORF.4"      "Ra2_promoter.1" "Ra2_promoter.2"
## [965] "Ra2_promoter.3" "sh2.5"          "sh2.6"          "sh2.7"
## [969] "sh2.9"          "su1.4"          "su1.5"          "su1.7"
## [973] "tb1.17"         "tb1.18"         "tb1.19"         "tb1.5"
## [977] "te1.3"          "te1.4"          "zagl1.1"        "zagl1.6"
## [981] "zap1.2"         "zen1.1"         "zen1.2"         "zen1.4"
## [985] "zfl2.6"         "zmm3.4"
```

```r
head(pos_raw)
```

```
## # A tibble: 6 x 15
##   SNP_ID   cdv_marker_id Chromosome Position  alt_pos mult_positions amplicon
##   <chr>    <chr>         <chr>      <chr>     <chr>   <chr>          <chr>
## 1 abph1.20 5976          2          27403404  <NA>    <NA>           abph1
## 2 abph1.22 5978          2          27403892  <NA>    <NA>           abph1
## 3 ae1.3    6605          5          167889790 <NA>    <NA>           ae1
## 4 ae1.4    6606          5          167889682 <NA>    <NA>           ae1
## 5 ae1.5    6607          5          167889821 <NA>    <NA>           ae1
## 6 an1.4    5982          1          240498509 <NA>    <NA>           an1
## # i 8 more variables: cdv_map_feature.name <chr>, gene <chr>,
## #   `candidate/random` <chr>, Genaissance_daa_id <chr>, Sequenom_daa_id <chr>,
## #   count_amplicons <chr>, count_cmf <chr>, count_gene <chr>
```

```r
tail(pos_raw)
```

```
## # A tibble: 6 x 15
##   SNP_ID cdv_marker_id Chromosome Position  alt_pos mult_positions amplicon
##   <chr>  <chr>         <chr>      <chr>     <chr>   <chr>          <chr>
## 1 zap1.2 3514          2          233128584 <NA>    <NA>           zap1
## 2 zen1.1 3519          unknown    unknown   <NA>    <NA>           zen1
## 3 zen1.2 3520          unknown    unknown   <NA>    <NA>           zen1
## 4 zen1.4 3521          unknown    unknown   <NA>    <NA>           zen1
```

```
## 5 zfl2.6 6463          2          12543294  <NA>    <NA>           zfl2
## 6 zmm3.4 3527          9          16966348  <NA>    <NA>           zmm3
## # i 8 more variables: cdv_map_feature.name <chr>, gene <chr>,
## #   `candidate/random` <chr>, Genaissance_daa_id <chr>, Sequenom_daa_id <chr>,
## #   count_amplicons <chr>, count_cmf <chr>, count_gene <chr>
```

```r
names(pos_raw)
```

```
##  [1] "SNP_ID"               "cdv_marker_id"        "Chromosome"
##  [4] "Position"             "alt_pos"              "mult_positions"
##  [7] "amplicon"             "cdv_map_feature.name" "gene"
## [10] "candidate/random"     "Genaissance_daa_id"   "Sequenom_daa_id"
## [13] "count_amplicons"      "count_cmf"            "count_gene"
```

Standardize key columns

- The first column of the genotype file is used as the sample identifier, and the third column is used as the group variable. From the SNP position file, I keep only the SNP ID, chromosome, and position columns. Output is a count of intersect of SNP ID.

```r
sample_col <- names(geno)[1]
group_col  <- names(geno)[3]

pos <- pos_raw %>%
  transmute(
    SNP_ID     = .data[[names(pos_raw)[1]]],
    Chromosome = .data[[names(pos_raw)[3]]],
    Position   = .data[[names(pos_raw)[4]]]
  )

# Check intersect of two files (SNP_ID)

snp_cols <- intersect(names(geno), pos$SNP_ID)

# Error check

if (length(snp_cols) == 0) {
  stop("No overlapping SNP columns were found between genotype data and position data.")
}

# Recheck new variable
length(snp_cols)
```

```
## [1] 983
```

Define maize and teosinte groups

```r
maize_groups    <- c("ZMMIL", "ZMMLR", "ZMMMR")
teosinte_groups <- c("ZMPBA", "ZMPIL", "ZMPJA")
```

Transpose the genotype data, standardize missing values, classify SNP position records, and write the final output files.

```r
# Convert genotype rows into SNP rows

make_genotype_by_snp <- function(geno_df, groups, sample_col, group_col, snp_cols) {
  sub_df <- geno_df %>%
    filter(.data[[group_col]] %in% groups)

  if (nrow(sub_df) == 0) {
    stop("No samples found for groups: ", paste(groups, collapse = ", "))
  }

  sample_ids <- make.unique(sub_df[[sample_col]])

  geno_matrix <- sub_df %>%
    select(all_of(snp_cols)) %>%
    as.matrix()

  geno_t <- as_tibble(t(geno_matrix), .name_repair = "minimal")
  names(geno_t) <- sample_ids

  geno_t %>%
    mutate(SNP_ID = snp_cols, .before = 1)
}

# Replace missing genotype values for output files

replace_missing_symbol <- function(df, symbol) {
  genotype_cols <- setdiff(names(df), c("SNP_ID", "Chromosome", "Position"))

  df %>%
    mutate(
      across(
        all_of(genotype_cols),
        ~ case_when(
          is.na(.x) ~ symbol,
          str_trim(.x) == "" ~ symbol,
          .x %in% c("NN", "N", "NA", "na") ~ symbol,
          TRUE ~ .x
        )
      )
    )
}

# Join positions to genotype data and classify known SNPs

prepare_joined_data <- function(geno_snp_df, pos_df) {

  pos_clean <- pos_df %>%
    mutate(
      chr_num = suppressWarnings(as.integer(Chromosome)),
      pos_num = suppressWarnings(as.numeric(Position))
```

```r
    )

  multiple_ids <- pos_clean %>%
    count(SNP_ID, name = "n_rows") %>%
    filter(n_rows > 1) %>%
    pull(SNP_ID)

  unknown_ids <- pos_clean %>%
    filter(!(SNP_ID %in% multiple_ids)) %>%
    filter(is.na(chr_num) | is.na(pos_num) | !(chr_num %in% 1:10)) %>%
    pull(SNP_ID) %>%
    unique()

  known_ids <- pos_clean %>%
    filter(!(SNP_ID %in% multiple_ids)) %>%
    filter(!(SNP_ID %in% unknown_ids)) %>%
    pull(SNP_ID) %>%
    unique()

  joined <- pos_clean %>%
    left_join(geno_snp_df, by = "SNP_ID") %>%
    select(SNP_ID, Chromosome, Position, everything(), -chr_num, -pos_num)

  list(
    known    = joined %>% filter(SNP_ID %in% known_ids),
    unknown  = joined %>% filter(SNP_ID %in% unknown_ids),
    multiple = joined %>% filter(SNP_ID %in% multiple_ids)
  )
}

# Write required output files

write_output_files <- function(data_list, out_subdir, prefix) {

  known <- data_list$known

  for (chr in 1:10) {
    chr_data <- known %>%
      filter(Chromosome == as.character(chr))

    asc <- chr_data %>%
      mutate(Position_numeric = suppressWarnings(as.numeric(Position))) %>%
      arrange(Position_numeric) %>%
      select(-Position_numeric) %>%
      replace_missing_symbol("?")

    desc <- chr_data %>%
      mutate(Position_numeric = suppressWarnings(as.numeric(Position))) %>%
      arrange(desc(Position_numeric)) %>%
      select(-Position_numeric) %>%
      replace_missing_symbol("-")

    asc_file  <- file.path(out_subdir, paste0(prefix, "_chr", chr, "_ascending.txt"))
```

```r
    desc_file <- file.path(out_subdir, paste0(prefix, "_chr", chr, "_descending.txt"))

    readr::write_tsv(asc, asc_file, na = "")
    readr::write_tsv(desc, desc_file, na = "")
  }
}
```

Build maize dataset

```r
maize_snp <- make_genotype_by_snp(
  geno_df    = geno,
  groups     = maize_groups,
  sample_col = sample_col,
  group_col  = group_col,
  snp_cols   = snp_cols
)

maize_data <- prepare_joined_data(maize_snp, pos)

write_output_files(
  data_list  = maize_data,
  out_subdir = file.path(out_dir, "maize"),
  prefix     = "maize"
)
```

Build teosinte dataset

```r
teosinte_snp <- make_genotype_by_snp(
  geno_df    = geno,
  groups     = teosinte_groups,
  sample_col = sample_col,
  group_col  = group_col,
  snp_cols   = snp_cols
)

teosinte_data <- prepare_joined_data(teosinte_snp, pos)

write_output_files(
  data_list  = teosinte_data,
  out_subdir = file.path(out_dir, "teosinte"),
  prefix     = "teosinte"
)
```

Check output summary

- The assignment requires 40 output files total: 20 for maize and 20 for teosinte.

```r
all_output_files <- list.files(out_dir, recursive = TRUE, full.names = TRUE)

cat("=== OUTPUT SUMMARY ===\n")
```

```
## === OUTPUT SUMMARY ===
```

```r
cat("Maize files:", length(list.files(file.path(out_dir, "maize"))), "\n")
```

```
## Maize files: 20
```

```r
cat("Teosinte files:", length(list.files(file.path(out_dir, "teosinte"))), "\n\n")
```

```
## Teosinte files: 20
```

```r
cat("Output directory:\n", normalizePath(out_dir), "\n")
```

```
## Output directory:
##  /Users/fan/BCB546_R_assignment/02_output_files
```

Part II. Visualization

Prepare data for plotting

- For the visualization step, I use only the known SNP positions and convert the wide genotype tables into long format. Each genotype is classified as homozygous, heterozygous, or missing.

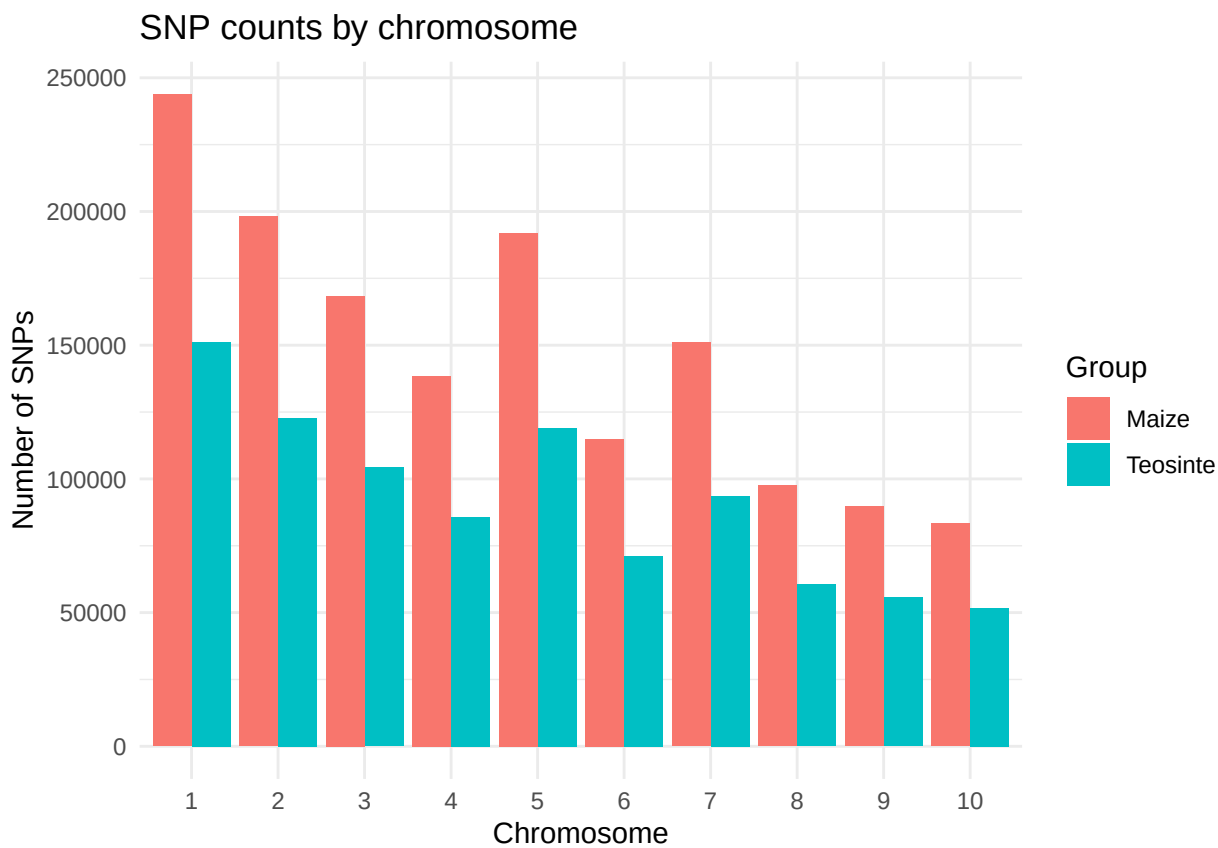
```r
# Create new directory
dir.create(file.path(out_dir, "figures"), showWarnings = FALSE, recursive = TRUE)

# Convert wide genotype table to long tidy format

maize_joined <- maize_data$known
teosinte_joined <- teosinte_data$known

make_long_genotype <- function(df, group_label) {

  genotype_cols <- setdiff(names(df), c("SNP_ID", "Chromosome", "Position"))

  df %>%
    pivot_longer(
      cols = all_of(genotype_cols),
      names_to = "SampleID",
      values_to = "Genotype"
    ) %>%
    mutate(
      Group = group_label,
      Genotype = str_squish(as.character(Genotype)),
      State = case_when(
        is.na(Genotype) | Genotype == "" | Genotype %in% c("NN", "N/N", "NA", "N", "?/?") ~ "Missing",
```

```r
        str_remove_all(Genotype, "/") %in% c("AA", "CC", "GG", "TT") ~ "Homozygous",
        TRUE ~ "Heterozygous"
      )
    )
}

maize_long <- make_long_genotype(maize_joined, "Maize")
teosinte_long <- make_long_genotype(teosinte_joined, "Teosinte")

all_long <- bind_rows(maize_long, teosinte_long) %>%
  mutate(
    Group = factor(Group, levels = c("Maize", "Teosinte")),
    State = factor(State, levels = c("Homozygous", "Heterozygous", "Missing")),
    Chromosome = factor(Chromosome, levels = as.character(1:10))
  )
```

Plot. I: SNP counts by chromosome

- This plot shows how SNPs are distributed across chromosomes in maize and teosinte. Because the long dataset contains one row per SNP per sample, I keep only distinct SNP IDs before counting.

```r
# Make chr into number order
snp_count_plot_data <- all_long %>%
  mutate(Chromosome = factor(Chromosome, levels = 1:10))
# Make a graph
ggplot(snp_count_plot_data, aes(x = Chromosome, fill = Group)) +
  geom_bar(position = "dodge") +
  labs(
    title = "SNP counts by chromosome",
    x = "Chromosome",
    y = "Number of SNPs"
  ) +
  theme_minimal()
```

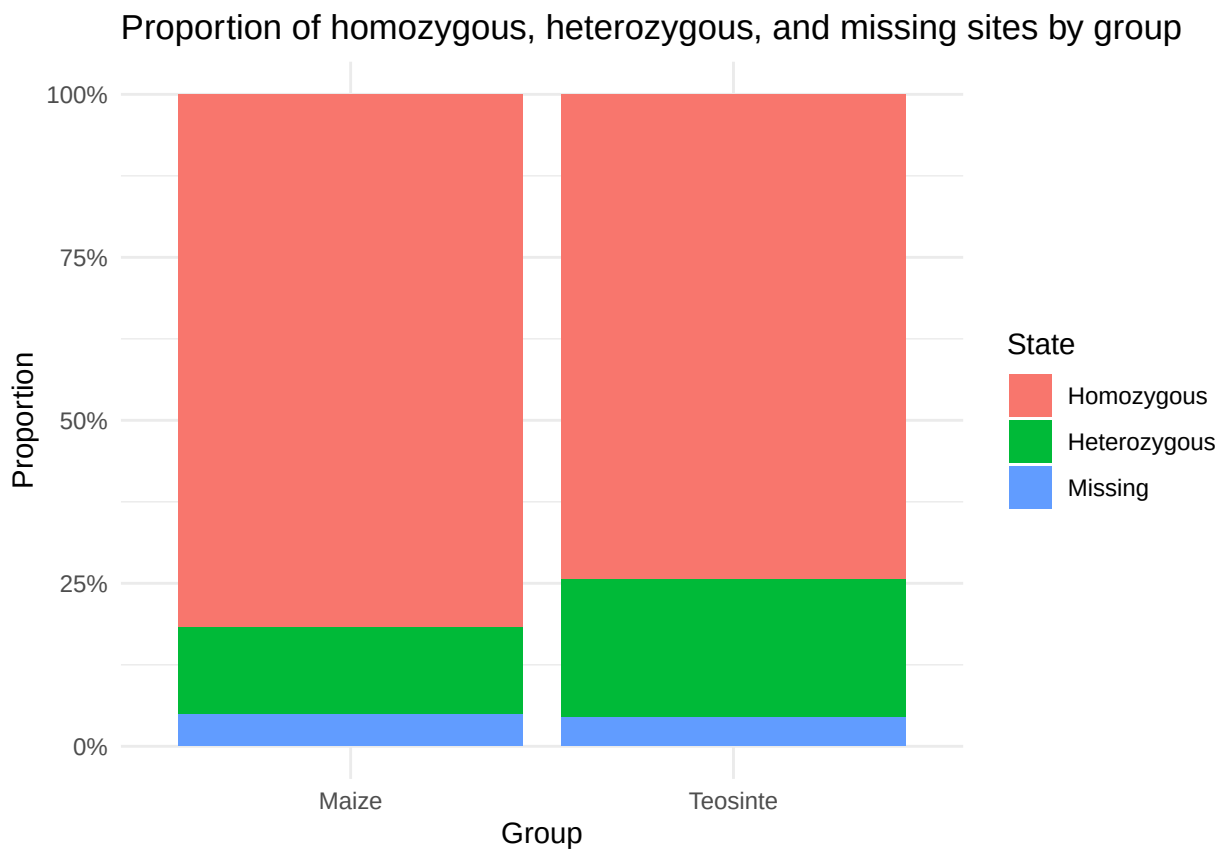



```r
# Save Plot
ggsave(
  file.path(out_dir, "figures", "plot_snp_counts_by_chromosome.png"),
  width = 8, height = 5, dpi = 300
)
```

Plot. II: Proportion of homozygous, heterozygous, and missing sites by group

- This plot summarizes genotype composition at the group level

```r
group_plot_data <- all_long %>%
  count(Group, State) %>%
  complete(Group, State, fill = list(n = 0))

ggplot(group_plot_data, aes(x = Group, y = n, fill = State)) +
  geom_col(position = "fill") +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    title = "Proportion of homozygous, heterozygous, and missing sites by group",
    x = "Group",
    y = "Proportion"
  ) +
  theme_minimal()
```



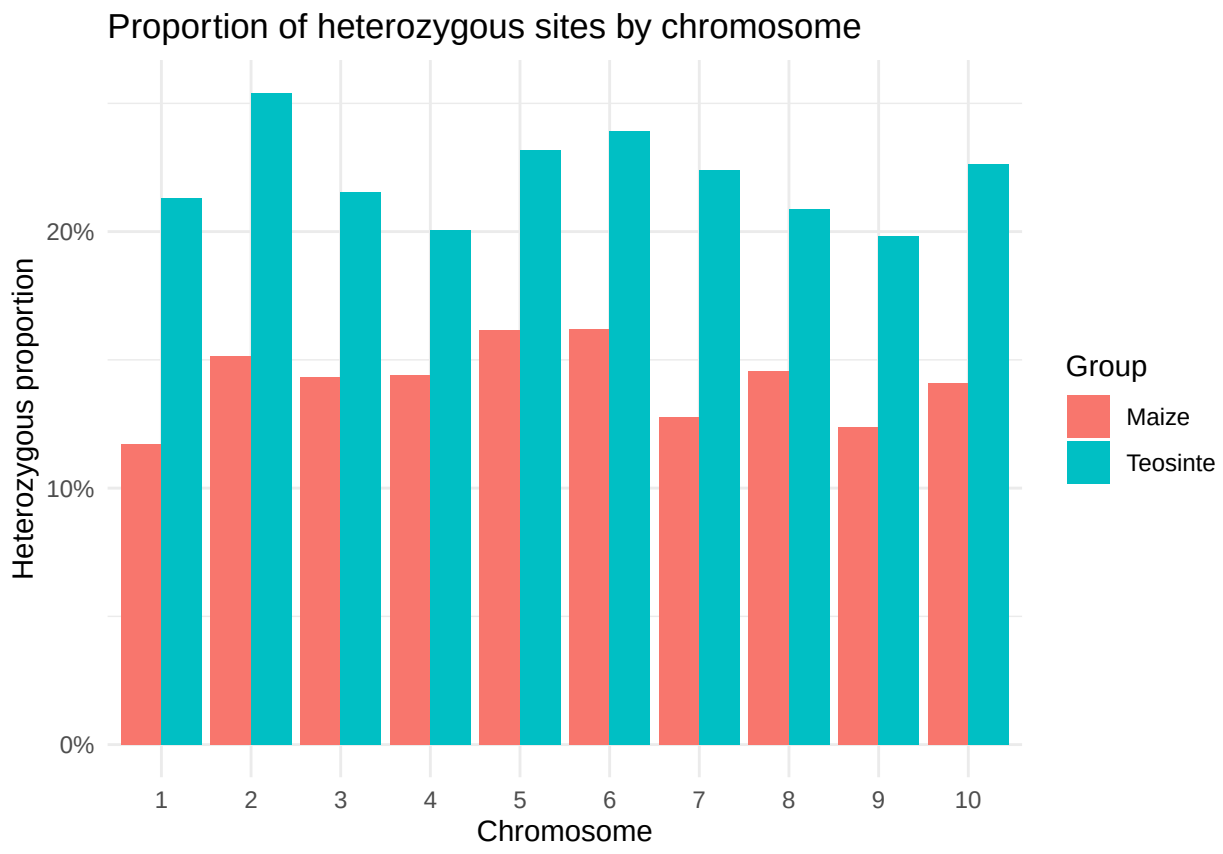

```
ggsave(
  file.path(out_dir, "figures", "plot_state_proportion_by_group.png"),
  width = 7, height = 5, dpi = 300
)
```

Plot. III: My own visualization: proportion of heterozygous sites by chromosome

- As an additional visualization, I plotted the proportion of heterozygous sites by chromosome for maize and teosinte.

```
chrom_het_data <- all_long %>%
  filter(State != "Missing") %>%
  group_by(Group, Chromosome) %>%
  summarise(
    heterozygous_prop = mean(State == "Heterozygous"),
    .groups = "drop"
  )

ggplot(chrom_het_data, aes(x = Chromosome, y = heterozygous_prop, fill = Group)) +
  geom_col(position = "dodge") +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    title = "Proportion of heterozygous sites by chromosome",
    x = "Chromosome",
```

```
    y = "Heterozygous proportion"
  ) +
  theme_minimal()
```




```
ggsave(
  file.path(out_dir, "figures", "plot_heterozygosity_by_chromosome.png"),
  width = 8, height = 5, dpi = 300
)
```